

PERM BOLSHOYE SAVINO, Russian Federation

WMO: 282250

Lat: 57.915N

Lon: 56.021E

Elev: 123

StdP: 99.85

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.8	(c) -27.2	(d) -34.0	(e) 0.2	(f) -30.6	(g) -30.8	(h) 0.2	(i) -27.4	(j) 11.3	(k) -7.4	(l) 10.3	(m) -7.3	(n) 2.0	(o) 240	(p) 0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.0	20.3	28.1	19.3	26.1	18.3	21.4	27.5	20.4	26.5	19.4	24.8	3.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.1	24.3	18.1	13.2	23.0	17.2	12.4	21.9	62.8	27.7	59.4	26.4	56.0	24.5	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.1	(b) 9.0	(c) 8.0	(e) -33.9	(f) 32.3	(g) 3.8	(h) 2.0	(i) -36.7	(j) 33.7	(k) -38.9	(l) 34.9	(m) -41.1	(n) 36.0	(o) -43.8	(p) 37.4		
(a) 10.1	(b) 9.0	(c) 8.0	(e) -34.3	(f) 22.8	(g) 3.7	(h) 1.3	(i) -37.0	(j) 23.7	(k) -39.2	(l) 24.5	(m) -41.3	(n) 25.3	(o) -44.0	(p) 26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-12.9	-12.3	-4.2	3.9	11.7	16.2	18.6	16.1	10.5	3.4	-5.0	-10.7
	DBStd	12.57	7.69	7.43	4.69	5.09	5.11	4.45	3.87	4.23	4.24	4.70	6.82	7.69
	HDD10.0	3365	711	624	439	192	43	5	1	2	44	211	449	642
	HDD18.3	5701	969	858	698	433	212	92	45	96	236	463	699	901
	CDD10.0	820	0	0	0	9	95	191	268	191	59	6	0	0
	CDD18.3	116	0	0	0	0	6	28	54	27	1	0	0	0
	CDH23.3	1193	0	0	0	2	94	272	559	252	13	0	0	0
	CDH26.7	334	0	0	0	0	16	69	168	80	1	0	0	0
Wind	WSAvg	4.0	4.2	4.0	4.5	4.2	4.0	3.7	3.1	3.3	3.6	4.3	4.3	4.2
Precipitation	PrecAvg	799	61	44	51	44	60	87	82	84	70	78	73	64
	PrecMax	1040	105	85	90	77	106	160	162	182	133	144	129	107
	PrecMin	632	14	4	5	1	23	37	17	24	15	35	19	26
	PrecStd	91	21	20	22	21	23	30	35	33	27	27	26	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	2.1	7.9	23.0	28.9	31.1	32.1	32.2	25.1	18.2	9.2	2.2
		MCWB	0.9	0.8	3.7	13.1	17.0	20.7	21.1	20.1	15.5	11.4	8.3	1.8
	2%	DB	0.2	1.0	5.0	18.2	26.0	28.5	30.2	28.9	22.0	15.0	6.8	1.2
		MCWB	-0.2	0.4	2.0	10.7	15.2	19.6	20.6	19.5	15.4	10.4	5.9	0.8
	5%	DB	-0.9	-0.1	3.2	15.1	23.8	26.8	28.8	26.0	19.8	12.0	4.0	0.7
		MCWB	-1.4	-0.8	1.1	8.9	14.4	18.8	20.1	18.8	14.2	8.9	3.4	0.3
	10%	DB	-2.9	-2.1	2.0	12.1	21.1	24.1	26.8	23.4	17.1	9.9	2.1	-1.1
		MCWB	-3.4	-2.9	0.6	6.9	13.1	17.3	19.3	17.8	13.1	8.2	1.5	-1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	1.0	4.1	13.5	18.9	22.3	22.8	21.7	17.5	12.7	8.5	2.1
		MCDB	1.2	1.3	6.8	20.8	25.7	28.8	29.1	28.0	21.7	16.3	9.2	2.1
	2%	WB	0.0	0.2	2.5	11.3	16.7	20.7	21.7	20.6	16.0	10.8	5.5	0.9
		MCDB	0.2	0.7	3.9	17.1	23.5	26.9	27.9	26.8	20.4	14.0	6.2	1.1
	5%	WB	-1.5	-1.1	1.5	9.5	15.3	19.4	20.8	19.4	14.9	9.4	3.3	-0.1
		MCDB	-1.1	-0.4	2.7	14.5	21.8	24.9	26.9	24.7	18.8	11.5	3.7	0.3
	10%	WB	-3.3	-3.0	0.6	7.5	14.0	18.2	19.8	18.2	13.7	8.1	1.5	-1.8
		MCDB	-2.9	-2.3	1.7	11.4	19.7	23.0	25.7	22.7	16.6	9.9	1.8	-1.4
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.7	8.0	9.7	11.5	10.5	10.7	9.6	8.2	5.4	4.9	5.9
		MCDBR	5.8	5.5	8.2	14.5	15.4	13.3	14.1	13.6	12.1	9.5	4.9	3.8
		MCWBR	5.8	5.0	6.0	8.0	7.3	6.3	6.2	6.2	6.6	5.9	4.5	4.0
	5% WB	MCDBR	5.8	5.5	6.7	13.5	13.2	12.1	12.6	12.3	10.4	8.3	4.8	4.0
		MCWBR	5.9	5.2	5.4	7.8	7.2	6.4	6.0	6.1	6.2	5.7	4.6	4.2
Clear-Sky Solar Irradiance	taub	0.293	0.320	0.375	0.405	0.395	0.391	0.414	0.428	0.380	0.350	0.311	0.289	
	taud	1.995	2.001	1.926	2.071	2.292	2.359	2.303	2.234	2.332	2.266	2.095	1.970	
	Ebn at Noon	500	675	736	793	838	847	814	768	752	655	482	375	
	Edh at Noon	60	94	135	138	119	113	117	117	91	73	54	45	
All-Sky Solar Radiation	RadAvg	0.53	1.41	2.70	4.00	5.10	5.43	5.45	4.07	2.40	1.03	0.43	0.29	
	RadStd	0.08	0.23	0.33	0.53	0.57	0.48	0.58	0.40	0.25	0.15	0.09	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page